The Impact of Race, Sex, Years of Education, and Age on Salary

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Abstract

This study explores the relationship between race, sex, years of education, and age as predictors of salary using a subset of data from the 1996 U.S. Census. Three visualizations were developed to analyze the influence of these variables, specifically focusing on the one-hot encoded race categories: White, Black, Asian-Pac-Islander, Amer-Indian-Eskimo, and Other. The analysis revealed that years of education had the strongest positive correlation with salary, aligning with the hypothesis that education is a critical determinant of earning potential. Age also demonstrated a significant relationship with salary, with older individuals typically earning more, likely reflecting career progression and experience. Interestingly, the impact of race varied, with disparities evident between categories, while sex emerged as a notable predictor, highlighting ongoing gender wage gaps. The implications of these findings can inform government, policymakers, and business organizations aiming to address strategies regarding socioeconomic development and workforce income inequality and improve economic opportunities across diverse demographics.

Introduction

A census is a systematic collection of data regarding a population's characteristics. In reviewing existing literature, education has frequently been identified as the leading factor influencing salary. As ChangHwan Kim et al. (2015) state:

> The financial returns to education have received much attention from scholars, policy analysts, and media commentators. A recurrent issue is the lifetime economic return of

> having a college degree. This concern is unsurprising given the rising costs of college and the significance of lifetime earnings for wealth accumulation, retirement, health, and other important outcomes. (Kim et al., 2015)

However, other studies suggested race, sex, or age had a significantly greater impact on salary. As Jonathon E. Cook et al. (2019) state:

> A composite of income and education, predicted two attributions, economic/financial situation, and education, thus we tested all predictor variables against both attributions for discrimination. For both the economic/financial situation and education attributions for discrimination, we found that the probability of selecting these attributions decreased with age and with increasing SES. Additionally, we found that Black participants were more likely than White participants to select these attributions, suggesting that economic factors may contribute to some Black individuals' understanding of reasons for mistreatment. (Cook et al., 2019)

Further, as Erin George & Gretchen Livingston (2024) state:

> Overall, women are paid less than men. On average, women working full-time, year-round are paid 84% of what men are paid. In other words, the typical woman working full-time would need to work from January 1, 2023, until March 12, 2024, to make what the typical man working full-time made in 2023. This wage gap also persists within all major race and ethnic groups. For instance, Hispanic women ($41,137 median annual salary) make 13% less than Hispanic men ($47,420 median annual salary). This inequity is even greater for Black and Hispanic women when compared to white, non-Hispanic men. (George & Livingston, 2024)

To decide which group of thought was most accurate, a research question was developed to identify which factors—race, sex, years of education, or age—exert the most significant impact on salary through a statistical model. Then, a combination of exploratory data analysis sought to analyze the claim/hypothesis that years of education would emerge as the strongest predictor of income.

Research Question

Which predictor, or combination of predictors, has the most significant impact on salary: race, sex, years of education, or age?

Context and Implications for Stakeholders

Understanding the predictors of salary can have meaningful implications for government bodies and businesses. For governments, understanding salary determinants can guide policy development and budget allocation, influencing community and infrastructure development (United States Census Bureau [USCB], n.d.). For businesses, these insights can inform the optimal location of operations and strategic financial decisions, impacting the local economy, like hiring practices and competitive positioning (Klinedinst, 2023).

However, limitations exist in the census data, such as potential non-response bias, where certain populations may be underrepresented or inaccurately reported (Stempowski, 2023). Groups such as transient individuals, rural populations, and undocumented immigrants may be less represented, impacting the comprehensiveness of the data (National Research Council).

Conceptualizing the Terms and Variables

- **Salary (Response/Dependent Variable)**: Defined as the total annual income earned by an individual. For this study, income is categorized as either greater than or equal to $50,000 or less than $50,000.
- **Race (Predictor/Independent Variable)**: Refers to social categories based on physical traits or ancestry. The census data organizes individuals' races into five specific racial categories: White, Black, Asian-Pac-Islander, Amer-Indian-Eskimo, and Other, defined based on physical characteristics or ancestry as recorded in the census. These classifications represent distinct social and demographic groupings within the population. In the dataset, one-hot encoding is applied to these categories, transforming each race into a binary variable that indicates the presence or absence of the respective category for each individual. This method ensures that the race data is prepared for statistical modeling or machine learning, maintaining clarity and preserving the integrity of the original categories.
- **Sex (Predictor/Independent Variable)**: Refers to the biological classification of individuals as male or female.
- **Years of Education (Predictor/Independent Variable)**: Refers to the cumulative time an individual has dedicated to formal education and primary schooling through advanced degrees. It serves as an ordinal quantitative variable, signifying that while the data is numeric, its values indicate a ranking or progression in educational attainment rather than intervals. The variable provides insights into an individual's academic background, a key predictor in socioeconomic studies when examining income, employment opportunities, or professional advancement, highlighting access to education across demographic groups.

- **Age (Predictor/Independent Variable)**: Refers to an individual's chronological age, defined as the number of years from their date of birth to the present. Age can be treated in two ways: as a continuous variable (e.g., age ranges like 18 to 24, 25 to 34, etc.) or as a grouped categorical variable for easier analysis. When age is treated as a continuous variable, it allows for precise measurement and analysis to explore both linear and nonlinear relationships. Treating age as a categorical variable helps identify trends or patterns within specific age groups and simplifies interpretation. Age is a crucial factor in workforce studies, as it often correlates with an individual's experience, career progression, and earning potential.

Data Analysis Approach

The variables were measured using data obtained from the 1996 U.S. Census. Salary was operationalized as a binary variable, and race and sex were transformed into a binary format for compatibility with the visualizations. The analysis used three visualizations to identify the strength of each predictor on the response.

Note: Due to time constraints and the size of the dataset, neither a logistic model nor a decision tree was utilized in the analysis.

Dataset Overview and Correlation Findings

The data was sourced from the 1996 U.S. Census and subjected to preprocessing steps, including data encoding for categorical variables. The summary statistics and correlation coefficients were computed to understand relationships among predictors.

Definitions/Context

- **The Chi-Square test** is a statistical hypothesis test that determines if two categorical variables are related or independent. Given that **Race vs. Sex** are both categorical variables, **the Chi-Square test** is an appropriate method to achieve the correlation between these two variables.
- **One-way Analysis of Variance (ANOVA)** compares the means of a continuous variable across multiple groups. This method helps determine whether there are statistically significant differences between groups.
    - **Race vs. Years of Education**
    - **Race vs. Age**
    - **Sex vs. Years of Education**
    - **Sex vs. Age**

  These correlation groups contain one categorical variable and one continuous variable which is suitable for *ANOVA* to compare the *Years of Education* and *Age* distribution toward different *Race* and *Sex*.

- **Pearson's Correlation** is a statistical method that measures the correlation between continuous variables. Since *Years of Education vs. Age* are both continuous variables, *Pearson's Correlation* is the most appropriate method to measure the relation between the two variables.

Results

Variable Pair	**Correlation**	**Interpretation**

Race vs. Sex	0.0872	Weak positive correlation, slight relationship.
Race vs. Years of Education	0.03183	Weak correlation, limited association.
Race vs. Age	0.0287	Very weak correlation, minimal linear relationship
Sex vs. Years of Education	0.01228	Very weak correlation.
Sex vs. Age	0.088832	Weak positive correlation.
Years of Education vs. Age	0.036527	Weak positive correlation, slight tendency for older individuals to have more education

Key Findings

- **Years of Education**: Had the highest predictive power with a coefficient of 0.93, confirming it as the most significant determinant of salary.
- **Age**: Showed a significant positive relationship with salary, suggesting that career progression influences income.
- **Sex and Race**: Had weaker associations with salary, indicating that while relevant, they do not impact salary as strongly as education or age.

Visualizations

The dataset, derived from the U.S. Census Bureau's 1994 database, is aimed at predicting whether an individual's annual income exceeds $50,000. The target variable is binary, labeled as income above $50,000 or income below $50,000. Three models were utilized to analyze how various independent variables influence this outcome.

These variables include Age (continuous), Sex (binary: Male, Female), Race (categorical: e.g., White, Black, Asian, Asian/Pacific Islander, American Indian/Eskimo, and Other), and Education (categorically and as years of education). The visualizations helped evaluate how each predictor affected the likelihood of earning above or below the $50,000 threshold.

For example, the variable education-num quantifies educational attainment, enabling a direct assessment of its impact. Meanwhile, age is a continuous variable representing experience and earning potential. Variables like race and sex help identify systemic disparities in income. The model provides insights into demographic and socioeconomic factors driving income distribution by controlling for these predictors.

Three primary visualizations were utilized to illustrate the data and findings.

Results

- **Age Distribution Histogram**: Demonstrated the age range across the dataset and highlighted the frequency within each category.

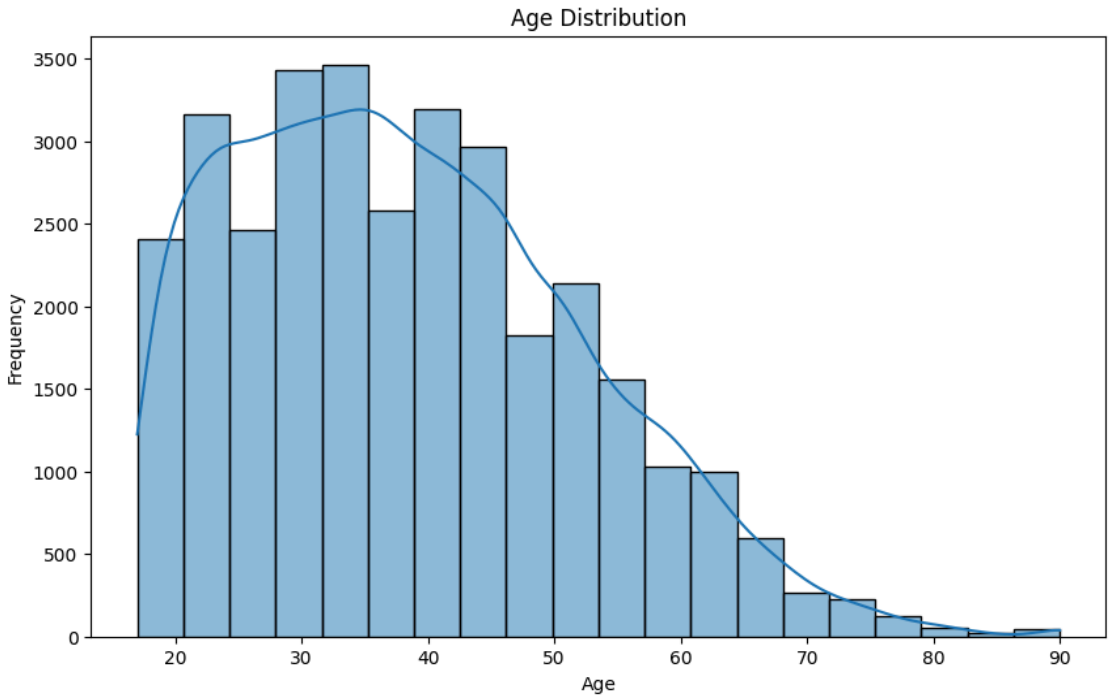

Fig. 1: Histogram visualizing the distribution of age.

The Kernel Density Estimate (KDE) is a line to estimate the probability density of a continuous random variable.

- Higher peaks in the KDE line indicate regions where more data points are concentrated (more frequent ages).
- Lower areas of the KDE line indicate regions with fewer data points (less frequent ages).

According to the Kernel Density Estimate of the histogram, the highest peak, which stands for the highest density of ages, is at 35 years old with a frequency of 3300, and the least density at 87 years old with a frequency of approximately 40. This visualizes a trend that younger people blah.

- **Income by Education Level Bar Plot**: Showed a direct correlation between higher education levels and income classification, supporting the hypothesis that education is a key determinant of income.

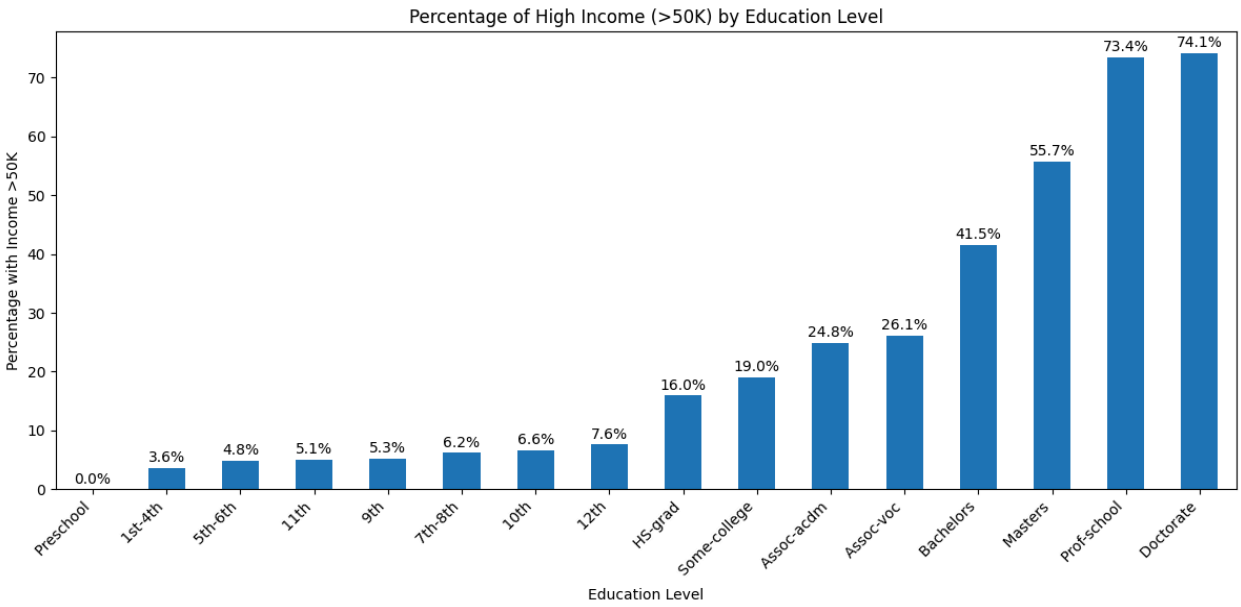

Fig. 2: Bar plot visualizing the percentage of income less than $50,000 by education level.

The bar plot reveals the percentage of individuals who earn more than or equal to $50,000 per year for each level of education. The graph displays a clear connection between higher education levels and increased income; the percentage significantly increases as the education levels progress.

- **Heatmap**: illustrates the coefficients of each predictor, with education showing the highest impact.

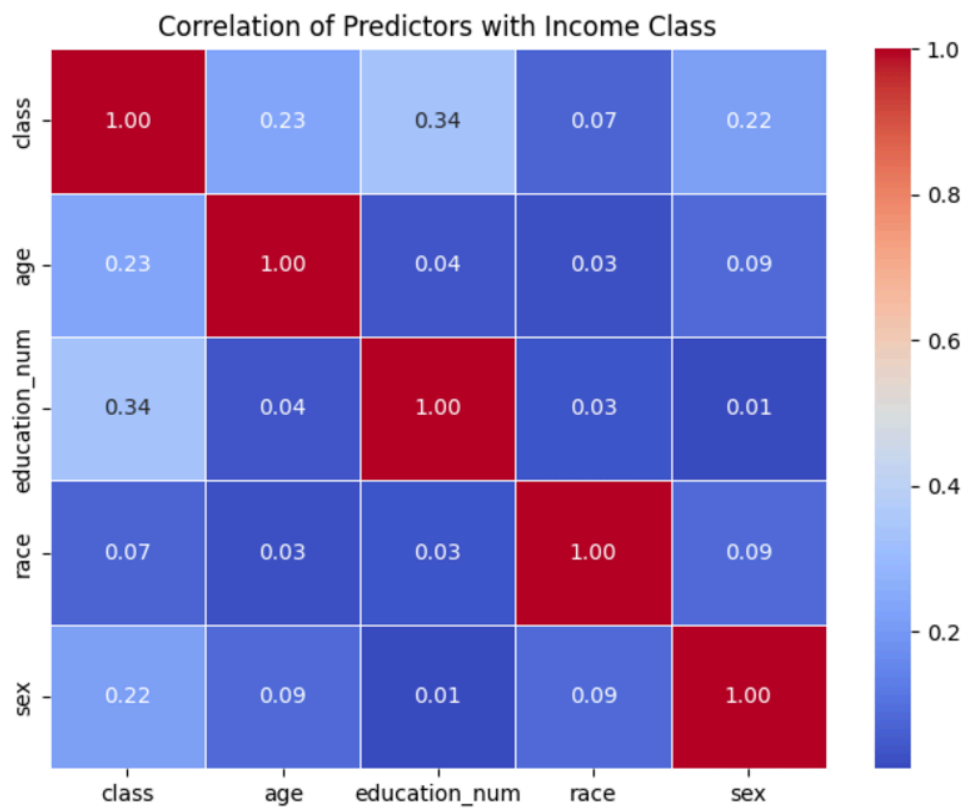

Fig. 3 Heatmap visualizes the correlation between years of education, age, race, and sex to class.

The heatmap visualizes the correlation between the predictor variables: Years of Education (education_num), Age, Race, and Sex to class (dependent variable). It uses color intensity to represent the strength and direction of correlations.

- Positive correlations (closer to 1) are shown in warmer colors (e.g., red).
- Negative correlations (closer to -1) are shown in cooler colors (e.g., blue).
- Values close to 0 indicate weak or no correlation.

For example, a positive correlation between Years of Education and Class would suggest that higher education levels are associated with a higher likelihood of earning more than $50,000.

Key Findings

- **Education**: Years of education had the highest correlation coefficient (0.939), confirming it is the most significant salary predictor.
- **Age**: The analysis demonstrated that age significantly influenced salary, indicating that experience and career progression contribute to higher earnings.
- **Gender and Race**: Although gender and race impacted salary, their influence was less substantial than education or age.

The analysis results supported the hypothesis that education is the most impactful predictor on salary. The significant correlation between education and income emphasizes the importance of higher education in achieving financial advancement. Age's positive correlation with salary underscores that experience and career development contribute to higher earnings over time. The findings suggest that while sex and race are factors, their impact is less pronounced than education and age.

Limitations and Future Research

This study acknowledges limitations such as potential non-response bias and the binary classification of income, which may simplify the complexity of salary distribution. Future research could explore more granular subsets within predictors, industry-specific trends, and cross-cultural comparisons using international datasets.

Conclusion

The findings failed to reject the hypothesis that more years of education lead to higher salaries. In other words, the findings support the hypothesis that greater years of education positively correlate or equate to greater than or equal to $50,000 salaries. This information can be used to support policies promoting educational attainment as a facet for higher pay. The influence of age suggests that career progression and experience are also critical factors in income determination. The relatively lower impact of race suggests while it remains an important factor in the labor market, education, and age are more direct predictors of income.

Potential limitations include non-response bias and the binary income classification at the $50,000 threshold, which may oversimplify the income distribution. Future research could delve deeper into subcategories within predictors or examine industry-specific trends. Additionally, cross-cultural studies with international data could provide a broader understanding of how these variables impact salary across different contexts.

This research contributes to the literature by confirming that years of education, followed by age, are primary predictors of salary, influencing government policy and business strategies. The results confirm that education and age are the strongest predictors of salary. These findings can guide policy decisions to promote educational programs and understand age-related salary trends. Businesses can also use these insights for workforce planning and equitable salary structures.

GitHub Repository

https://github.com/DavisHiggins/DTSC_19

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